# Skills Summary

## Geometric Series, $p$-Series, and the $n$th-Term Test

*Use this reference while completing your worksheet or when studying.*

---

### Skill 1 — Geometric series: classify by $r$, then sum

**Shape:** $\sum ar^k$ — every term is a fixed multiple $r$ of the one before it. Find $r$ by dividing any term by its predecessor.

**Test:** converges exactly when $|r| < 1$; diverges when $|r| \geq 1$. The sign of $r$ only makes the terms alternate.

**Sum (only when $|r| < 1$):** $S = \dfrac{a}{1-r}$, where $a$ is the *first term actually written*, not the coefficient. Substitute the starting index to find it.

**Example:** Find $\displaystyle\sum_{n=0}^{\infty} 5\left(\tfrac{1}{3}\right)^n$.

**Step 1.** Consecutive terms differ by the fixed factor $\frac{1}{3}$, so this is geometric and the sum formula applies once $|r| < 1$ is confirmed.

**Step 2.** $a = 5\left(\frac{1}{3}\right)^0 = 5$ and $r = \frac{1}{3}$, so $S = \dfrac{5}{1-\frac{1}{3}} = \dfrac{5}{\frac{2}{3}}$

$\Rightarrow \quad S = \mathbf{15/2}$

**Try it:** Find $\displaystyle\sum_{n=1}^{\infty} 8\left(\tfrac{1}{2}\right)^{n-1}$.

check: $S = \mathbf{16}$

## Skill 2 — $p$-series: classify by $p$ alone

**Shape:** $\sum \dfrac{1}{n^p}$ with $p > 0$ constant. Rewrite roots as powers first: $\sqrt{n} = n^{1/2}$, $\sqrt[3]{n^2} = n^{2/3}$.

**Test:** converges when $p > 1$; diverges when $p \le 1$. That single comparison is the whole test.

**The borderline case is** $p = 1$, the harmonic series $\sum \frac{1}{n}$, and it *diverges*. The test gives no sum — only a verdict.

**Example:** Classify $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{4/3}}$.

**Step 1.** The terms are a constant power of $n$ in the denominator, so this is a $p$-series and the only thing to decide is where $p$ sits relative to 1.

**Step 2.** Here $p = \frac{4}{3}$, and $\frac{4}{3} = 1.333\ldots$ is greater than 1

$\Rightarrow$ $\mathbf{4/3 > 1}$, so the series converges

**Try it:** Classify $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{2/5}}$ — write the comparison of $p$ with 1.

check: $\mathbf{2/5 < 1}$, diverges

## Skill 3 — The $n$th-term test: a divergence detector only

**Statement:** if $\displaystyle\lim_{n\to\infty} a_n \neq 0$ (or the limit fails to exist), then $\sum a_n$ **diverges**.

**What it never does:** a limit of 0 proves nothing at all. Say "inconclusive" and reach for another test.

**Doing the limit:** for a ratio of polynomials, compare degrees — equal degrees give the ratio of leading coefficients (nonzero, so the series diverges), and a bigger denominator degree gives 0 (inconclusive).

**Example:** Apply the $n$th-term test to $\displaystyle\sum_{n=1}^{\infty} \frac{4n+3}{n+7}$.

**Step 1.** Nothing here is geometric or a $p$-series, so start with the cheapest test — look at what the terms themselves approach.

**Step 2.** Numerator and denominator are both degree 1, so divide by $n$: $\displaystyle\lim_{n\to\infty} \frac{4+3/n}{1+7/n} = \frac{4}{1}$

$\Rightarrow$ $\lim a_n = \mathbf{4} \neq 0$, so the series diverges

**Try it:** Apply the $n$th-term test to $\displaystyle\sum_{n=1}^{\infty} \frac{5n^2-2}{2n^2+n}$ — find $\lim a_n$.

check: $\lim a_n = \mathbf{5/2}$, diverges

**(!) Watch out:** "the terms go to zero, so it converges" is the single most expensive error in this unit. It is the $n$th-term test read backwards. Both $\sum \frac{1}{n}$ (divergent) and $\sum \left(\frac{1}{2}\right)^n$ (convergent) have terms going to zero — the limit alone cannot tell them apart.